

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
APPLIED BUSINESS ANALYTICS
Practical –IX

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Practical 9: Hypothesis Testing and A/B Testing**Part A: Hypothesis Testing Dataset: supermarket_sales.csv****1. Load the supermarket_sales.csv dataset.**

```
import pandas as pd
import numpy as np
from scipy.stats import ttest_1samp
# Load dataset
df = pd.read_csv("supermarket_sales.csv")
[10] ✓ 0.0s
```

2. Display the first five records.

```
First five records:
Invoice ID Branch City Customer type Gender \
0 750-67-8428 A Yangon Member Female
1 226-31-3081 C Naypyitaw Normal Female
2 631-41-3108 A Yangon Normal Male
3 123-19-1176 A Yangon Member Male
4 373-73-7910 A Yangon Normal Male

Product line Unit price Quantity Tax % Total Date \
0 Health and beauty 74.69 7 26.1415 548.9715 1/5/2019
1 Electronic accessories 15.28 5 3.8200 80.2200 3/8/2019
2 Home and lifestyle 46.33 7 16.2155 340.5255 3/3/2019
3 Health and beauty 58.22 8 23.2880 489.0480 1/27/2019
4 Sports and travel 86.31 7 30.2085 634.3785 2/8/2019

Time Payment cogs gross margin percentage gross income Rating
0 13:08 Ewallet 522.83 4.761905 26.1415 9.1
1 10:29 Cash 76.40 4.761905 3.8200 9.6
2 13:23 Credit card 324.31 4.761905 16.2155 7.4
3 20:33 Ewallet 465.76 4.761905 23.2880 8.4
4 10:37 Ewallet 604.17 4.761905 30.2085 5.3
```

```
print("First five records:")
print(df.head())
[11] ✓ 0.0s
```

3. Select the revenue column.

```
Revenue column:
0 548.9715
1 80.2200
2 340.5255
3 489.0480
4 634.3785
...
995 42.3675
996 1022.4900
997 33.4320
998 69.1110
999 649.2990
Name: Total, Length: 1000, dtype: float64
```

```
# In this dataset, Total represents the sales/revenue amount
revenue = df["Total"]
print("Revenue column:")
print(revenue)
[12] ✓ 0.0s
```

4. Check for missing values.

```
print("Missing values in revenue:")
print(revenue.isnull().sum())
[13] ✓ 0.0s ... Missing values in revenue:
0
```

5. Calculate the mean revenue.

```
mean_revenue = revenue.mean()
print("Mean Revenue:", mean_revenue)
[14] ✓ 0.0s ... Mean Revenue: 322.966749
```

6. Assume that the expected average revenue is ₹1,500.

```
expected_revenue = 1500
print("Expected Average Revenue: ₹", expected_revenue)
[15] ✓ 0.0s ... Expected Average Revenue: ₹ 1500
```

7. Write the null hypothesis: $H_0: \mu = 1500$

```
# H0:  $\mu = 1500$ 
print("Null Hypothesis ( $H_0$ ):  $\mu = 1500$ ")
[17] ✓ 0.0s ... Null Hypothesis ( $H_0$ ):  $\mu = 1500$ 
```

8. Write the alternative hypothesis: $H_1: \mu \neq 1500$

```
# H1:  $\mu \neq 1500$ 
print("Alternative Hypothesis ( $H_1$ ):  $\mu \neq 1500$ ")
[18] ✓ 0.0s ... Alternative Hypothesis ( $H_1$ ):  $\mu \neq 1500$ 
```

9. Set the significance level as 5%.

```
alpha = 0.05
print("Significance Level:", alpha)
[19] ✓ 0.0s ... Significance Level: 0.05
```

10. Perform a one-sample t-test.

```

t_statistic, p_value = ttest_1samp(
    revenue,
    expected_revenue
)
print("T-statistic:", t_statistic)

```

... T-statistic: -151.37567896776733

11. Calculate the p-value.

```

print("P-value:", p_value)

```

... P-value: 0.0

12. Compare the p-value with 0.05.

```

if p_value < alpha:
    print("P-value is less than 0.05")
else:
    print("P-value is greater than or equal to 0.05")

```

... P-value is less than 0.05

13. State whether the null hypothesis is rejected or not rejected.

```

if p_value < alpha:
    print("Decision: Reject the Null Hypothesis (H0)")
else:
    print("Decision: Do Not Reject the Null Hypothesis (H0)")

```

... Decision: Reject the Null Hypothesis (H0)

14. Write the conclusion in simple business terms.

```

if p_value < alpha:
    print("Conclusion:")
    print("The average revenue is significantly different from ₹1,500.")
else:
    print("Conclusion:")
    print("There is not enough evidence to conclude that the average revenue is different from ₹1,500.")

```

... Conclusion:
The average revenue is significantly different from ₹1,500.

Part B: A/B Testing Dataset: ab_testing.csv**1. Load the A/B testing dataset.**

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.stats.proportion import proportions_ztest
df = pd.read_csv("ab_testing.csv")

```

2. Display the data.

```

print(df)

```

	Group	Users	Conversions
0	A	1000	120
1	B	1000	150

3. Calculate the conversion rate for Group A.

```

group_A = df[df["Group"] == "A"]
users_A = group_A["Users"].iloc[0]
conversions_A = group_A["Conversions"].iloc[0]
conversion_rate_A = conversions_A / users_A
print("Conversion Rate of Group A:", conversion_rate_A)
print("Conversion Rate of Group A:", conversion_rate_A * 100, "%")

```

... Conversion Rate of Group A: 0.12
Conversion Rate of Group A: 12.0 %

4. Calculate the conversion rate for Group B.

```

group_B = df[df["Group"] == "B"]
users_B = group_B["Users"].iloc[0]
conversions_B = group_B["Conversions"].iloc[0]
conversion_rate_B = conversions_B / users_B
print("Conversion Rate of Group B:", conversion_rate_B)
print("Conversion Rate of Group B:", conversion_rate_B * 100, "%")

```

... Conversion Rate of Group B: 0.15
Conversion Rate of Group B: 15.0 %

5. Compare the conversion rates of A and B.

```

print("Conversion Rate of Group A:", conversion_rate_A * 100, "%")
print("Conversion Rate of Group B:", conversion_rate_B * 100, "%")
difference = (conversion_rate_B - conversion_rate_A) * 100
print("Difference:", difference, "percentage points")

```

... Conversion Rate of Group A: 12.0 %
Conversion Rate of Group B: 15.0 %
Difference: 3.0 percentage points

6. Write the null hypothesis: $H_0: p_A = p_B$

```

print("H0: pA = pB")

```

... H0: pA = pB

7. Write the alternative hypothesis: $H_1: p_A \neq p_B$

```

print("H1: pA != pB")

```

... H1: pA != pB

8. Set the significance level as 5%.

```
alpha = 0.05
print("Significance Level:", alpha)
```

... Significance Level: 0.05

9. Perform a two-proportion test.

```
conversions = np.array([conversions_A, conversions_B])
users = np.array([users_A, users_B])
z_statistic, p_value = proportions_ztest(
    conversions,
    users
)
print("Z-statistic:", z_statistic)
```

... Z-statistic: -1.963049807622355

10. Calculate the p-value.

```
print("P-value:", p_value)
```

... P-value: 0.04964038686536878

11. Compare the p-value with 0.05.

```
if p_value < 0.05:
    print("P-value is less than 0.05")
else:
    print("P-value is greater than or equal to 0.05")
```

... P-value is less than 0.05

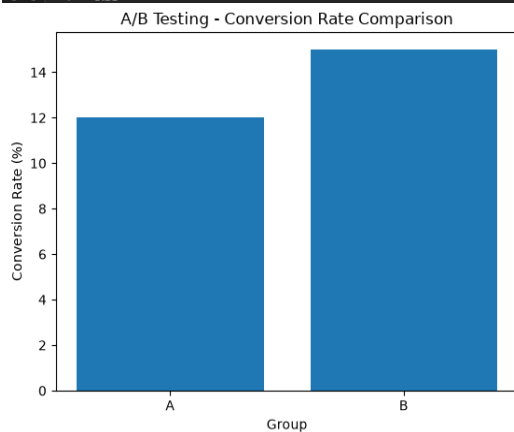
12. State whether the null hypothesis is rejected or not rejected.

```
if p_value < 0.05:
    print("Reject the Null Hypothesis (H0)")
else:
    print("Do Not Reject the Null Hypothesis (H0)")
```

... Reject the Null Hypothesis (H0)

13. Create a bar chart to compare the conversion rates.

```
groups = ["A", "B"]
conversion_rates = [
    conversion_rate_A * 100,
    conversion_rate_B * 100
]
plt.bar(groups, conversion_rates)
plt.xlabel("Group")
plt.ylabel("Conversion Rate (%)")
plt.title("A/B Testing - Conversion Rate Comparison")
plt.show()
```

**14. Write the conclusion based on the test result.**

```
if p_value < 0.05:
    print("Conclusion:")
    print("There is a statistically significant difference")
    print("between the conversion rates of Group A and Group B.")
else:
    print("Conclusion:")
    print("There is no statistically significant difference")
    print("between the conversion rates of Group A and Group B.")
```

... Conclusion:
There is a statistically significant difference
between the conversion rates of Group A and Group B.